



Technical Service Bulletin

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TELC Plus Silicates in Cummins Engines

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

For Cummins engines that are using ChevronTexaco® or Shell® Rotella Extended Life Coolant (TELC) Plus Silicates

Cummins recommends Heavy Duty engines and High Horsepower engines using ChevronTexaco, Shell Rotella, and their private label counterpart Extended Life OAT Coolants, which do **not** meet the elastomer compatibility section of CES 14603 are acceptable for extended service interval use assuming the initial coolant fill requirements were met from the vehicles' original equipment manufacturer (OEM).

Heavy Duty and High Horsepower engine overhauls or repairs involving the replacement of the following components, using this OAT coolant **must** discard the coolant and replace with new coolant.

- rocker lever housing gasket
- oil cooler housing gasket
- cylinder head gasket
- thermostat housing gasket.

If the replacement coolant is ChevronTexaco, Shell Rotella, or their private label counterpart Extended Life OAT Coolants, which do **not** meet the elastomer compatibility section of CES 14603, then the coolant **must** be treated by adding one 8 ounce bottle of liquid silicate fluid for every 12 gallons of total coolant system volume. It is critical to **not** overtreat the coolant with silicate fluid.

To obtain order forms or ask questions relative to ordering the silicate fluid:

Phone: 1-800-346-9041

Fax: 1-800-876-5317

Mail To: Silicate Fluid Order Program

PO Box 27388

Houston, TX 77227-7388

For technical questions on the Silicate Fluid, please call 1-800-782-7852 and choose option 3.

Additional important information is included in Figure 1 (Global Customer Bulletin) and Figure 2 (Frequently Asked Questions).

If there are any additional engine-related questions, please call Cummins Technical Hotline at 1-800-DIESELS.

Figure 1:

Global Customer Bulletin

This communication is being sent to update customers of Cummins Inc., ChevronTexaco, Shell Rotella, and Equilon® Lubricants on the issue of gasket coolant leakage in Cummins N14 engines, when factor filled with ChevronTexaco, Shell Rotella Extended Life OAT Coolants.

As stated in previous Customer Bulletins, Cummins, ChevronTexaco, Shell Rotella, and Equilon have been working together to find solutions.

Cummins has approved ChevronTexaco, Shell Rotella Extended Life OAT Coolants as a factory fill antifreeze/coolant in the following Cummins engine worldwide:

ISB, ISB ^e and B series engines

ISC and C series engines

ISL series engines

Cummins has approved ChevronTexaco, Shell Rotella Extended Life OAT Coolants plus a one-time addition of silicate as a factory fill antifreeze/coolant in the following Cummins engines worldwide:

N14 and N14 Plus engines

ISX and Signature engines

ISM, M11 and M11 Plus engines

High Horsepower Engines

The use of silicates as a preventive maintenance recommendation for the ISM, ISX engines and High Horsepower engines.

Please contact your local Equilon or ChevronTexaco, Shell representative for details on obtaining the proper silicate product and dosage.

This coolant may be used in off-road engines analogous to their over-the-road counterparts as outlined above.

We will jointly collaborate to determine what additional improvements might be possible, and will communicate the results of this work should new information become available.

Figure 2: Frequently Asked Questions

Q: I have a C (or B) series Cummins engine which was factory filled with ChevronTexaco, Shell Rotella Extended Life OAT Coolant. Do I need to add 'Silicate Fluid for Cummins Engines **only**' ('Silicate Fluid') to my coolant system?

A: No.

Q: I have a C (or B) series Cummins engine which was factory filled with ChevronTexaco, Shell Rotella Extended Life OAT Coolant. I have already added Silicate Fluid to my cooling system. Should I drain and flush now that I find that the addition of silicates was **not** necessary?

A: No.

Q: I have already added Silicate Fluid to my M11 which was factory filled with ChevronTexaco, Shell Rotella Extended Life OAT Coolant. Should I add more Silicate Fluid?

A: No.

Q: I have an N14 engine which was factory filled with ChevronTexaco, Shell Rotella Extended Life OAT Coolant and a rocker housing gasket is leaking. Can I use ChevronTexaco, Shell Rotella Extended Life OAT Coolant after I have the gasket replaced?

Yes. Just drain and dispose of the used coolant. Replace the rocker housing gasket and refill the cooling system with ChevronTexaco, Shell Rotella Extended Life OAT Coolant 50/50. Add one 8 ounce bottle of silicate fluid for every 12 gallons of total coolant system volume. Circulate the engine coolant for 15 to 20 minutes, or until the thermostat opens. Check the freeze point and adjust to -37°C [-34°F].

Q: I have an ISX which was factory filled with ChevronTexaco, Shell Rotella Extended Life OAT Coolants. I have already added Silicate Fluid to the cooling system. My coolant inadvertently became contaminated and I plan to flush and refill with new ChevronTexaco, Shell Rotella Extended Life OAT Coolant. Do I need to treat again with Silicate Fluid?

A: No. If the original gaskets remained in place, adding Silicate Fluid again will **not** be required.

Q: I have heard that using ChevronTexaco, Shell Rotella Extended Life OAT Coolant in Cummins engine was responsible for rocker housing gasket failures. Is this true?

A: A joint team of Cummins, Equilon, and ChevronTexaco scientists and engineers determined that a unique combination of circumstances needed to be present in order for the rocker housing gasket seepage to occur.

Q: I have heard that the use of ChevronTexaco, Shell Rotella Extended Life OAT Coolant can lead to catastrophic failure, including oil solidification (gelation).

A: After more than a year of intensive cooperative testing, Cummins has approved the use of ChevronTexaco, Shell Rotella Extended Life OAT Coolant in all Mid-Range, Heavy Duty, and High Horsepower engines (either with or without the one-time addition of Silicate Fluid, depending on the engine model, at the time of the factory fill). Cummins would **not** provide this endorsement if there was any concern that the use of ChevronTexaco, Shell Rotella Extended Life OAT Coolant would lead to catastrophic failure. Oil and coolant **must never** be mixed, regardless of the engine make or coolant formulation. The presence of coolant in the engine oil represents a serious situation that **must** be remedied immediately. Oil testing is the best way to determine if coolant

has mixed with the engine oil. Coolant intrusion into crankcase oil can lead to sludge, solidification, and potential engine failure, regardless of the coolant formulation being used. This includes conventional, fully formulated, extended service, and extended life coolants.

Q: My Cummins N14 engine was factory filled with ChevronTexaco, Shell Rotella Extended Life OAT Coolant. If I have a rocker housing gasket leak, will my Cummins warranty cover the repair?

A: Yes. After the rocker gasket is repaired, if you choose to continue using ChevronTexaco, Shell Rotella Extended Life OAT Coolant, be sure the repairing shop adds the Silicate Fluid and new 50/50 TELC to your cooling system.

Q: What has changed? Why is it okay to use the coolant now when it was once **not** okay?

A: Intense investigations proved that more than one factor caused the failure of a certain percentage of rocker housing gaskets in Cummins N14 engines. The addition of the Silicate Fluid is a preventive maintenance step recommended by ChevronTexaco, Equilon, and Cummins to reduce the possibility of seal degradation.

Q: How does one application of the silicate additive provide the required protection for engines such as the N14?

A: The effectiveness of silicates has been demonstrated in the laboratory, dyno, and field studies, and has been shown to be effective in extending the life of seals in Cummins engines at the recommended dosage level. It is believed that silicates create a protective barrier at the surface of the seal that does **not** allow free contact of coolant and the seal surface.

Q: Can ChevronTexaco, Shell Rotella Extended Life OAT Coolants be used with Cummins High-Horsepower industrial engines such as the QSK/KV, the V28, or the QSK78?

A: Yes. ChevronTexaco, Shell Rotella Extended Life OAT Coolants, which do **not** meet the elastometer compatibility section of CES 14603 are acceptable for extended service interval use in High-Horsepower industrial engines, assuming the initial coolant fill requirements were met from the original equipment manufacturer (OEM).

Q: My engine, which has ChevronTexaco, Shell Rotella Extended Life OAT Coolants Plus Silicates as coolant is being repaired. Assuming the coolant is **not** contaminated and in good condition, can it be reused?

A: The coolant can be reused if the repair procedure does **not** require the replacement of any cooling system gaskets. If any cooling system gaskets are replaced, the coolant **must** be replaced with new 50/50 TELC and one 8 ounce bottle of silicate fluid per every 12 gallons of total cooling system volume.

Q: How can I get more information on the maintenance requirements of ChevronTexaco, Shell Rotella Extended Life OAT Coolants and ChevronTexaco, Shell Rotella Extended Life OAT Coolants Plus Silicates? What tests are required to be sure the coolant's condition and operation meet the engine's needs?

A: Call 1-800-782-7852 and choose option 3 for technical information. If you would like to test TELC for inhibitor level at your site, use the Extended Life Coolant Test Strip (ELCTS). These test kits can be ordered from Analyst Inc. at 1-800-336-3637.

Note : It is very important to maintain freeze protection of the coolant. The optimum freeze protection is -37°C [-34°F] even if the engine is located in warm-weather climates.

Q: If an N14 rocker housing gasket is being replaced, will silicates need to be added regardless of mileage or hours since silicates were originally introduced to the engine?

A: Yes. It is recommended that the coolant be replaced with ChevronTexaco, Shell Rotella Extended Life OAT Coolants, 50/50, and one 8 ounce bottle of silicate fluid per every 12 gallons of total cooling system volume.

Q: Can excessive amounts of silicate fluid in the coolant induce issues that could impact engine performance?

A: If overtreatment is suspected, drain and flush the cooling system. Fill the cooling system with new coolant and add one 8 ounce bottle of silicate fluid per every 12 gallons of total cooling system volume.

Related Procedures

101-018-004 (/qs3/pubsys2/xml/en/procedures/101/101-018-004.html)	Coolant Recommendations and Specifications (/qs3/pubsys2/xml/en/procedures/101/101-018-004.html)
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